

The Embodiment Shortcut: Action-Following Metrics

Read the Robot, Not the Physics

Anonymous Authors

Abstract

Video world models for robotics are often evaluated by *action following*: an inverse dynamics model (IDM) infers the action from generated video and is compared with the action the generation was conditioned on. A low error is read as evidence that the model gets the interaction right. We show that, when the action is an actuator command, the manipulator’s pose is a near-direct readout of it, and trained IDMs read the arm and ignore the object. In a controlled MuJoCo push task with a position-controlled arm and a pre-registered analysis, deleting the object’s physics (placing it outside every reachable push path) changes the standard metric’s error by $0.15\times$ its reference error (95% CI [0.12, 0.19]): the object accounts for 0.3% of the metric’s distance to chance. We propose **OG-AF** (Object-Grounded Action-Following), which scores the frame pair at a delayed horizon after the arm has retracted, where the object accounts for 98%. In exploratory analyses OG-AF responds to friction changes that shift the object’s resting position by a few millimetres, while the standard metric stays flat across a 16-fold friction range. In a separately pre-registered world-model test, OG-AF separates a model trained on wrong physics from a correct one with paired effect $d = 1.94$ versus 0.46 for the standard metric, whose response is matched by its response on a no-object control.

1 Introduction

A video world model conditioned on a robot action should produce video in which that action has its physical consequences [4, 13, 18, 21]. *Action following* has become a common way to check this [5, 17]: train an IDM on real video, as in VPT-style labelling [1], run it on generated video, and measure how well it recovers the conditioning action. A low error is taken as evidence that the model follows the action, and implicitly that it gets the interaction right.

This inference fails for a structural reason. When the action is an actuator command and the arm is position-controlled, the arm’s configuration after executing the command is close to an invertible function of it. An IDM trained to minimise action-prediction error can recover the action from the arm alone, and the object then carries no additional information. The metric then scores how faithfully a model renders the robot, while being largely blind to whether contact physics was correct. We call this the *embodiment shortcut*: a shortcut in the sense of Geirhos et al. [8], and the evaluation-side analogue of causal confusion in imitation learning [6].

The question is not whether the arm *can* carry the action (in our setup it does by construction) but whether trained IDMs nonetheless use the object, and whether a simple change of horizon makes the metric physics-sensitive. We find that they do not, and that it does.

Contributions. (1) A controlled testbed that isolates the shortcut: paired rollouts in which the object is in the push path, out of it, or absent, with identical arm motion, and a paired estimator (§2, Fig. 1). (2) Pre-registered evidence that the standard metric is insensitive to object physics and that OG-AF is not (§3). (3) Exploratory evidence on how large a physics change each metric registers (§4). (4) A pre-registered world-model test in which OG-AF, but not the standard metric, separates a model trained on wrong physics, including a diagnosed failure of our first attempt (§5).

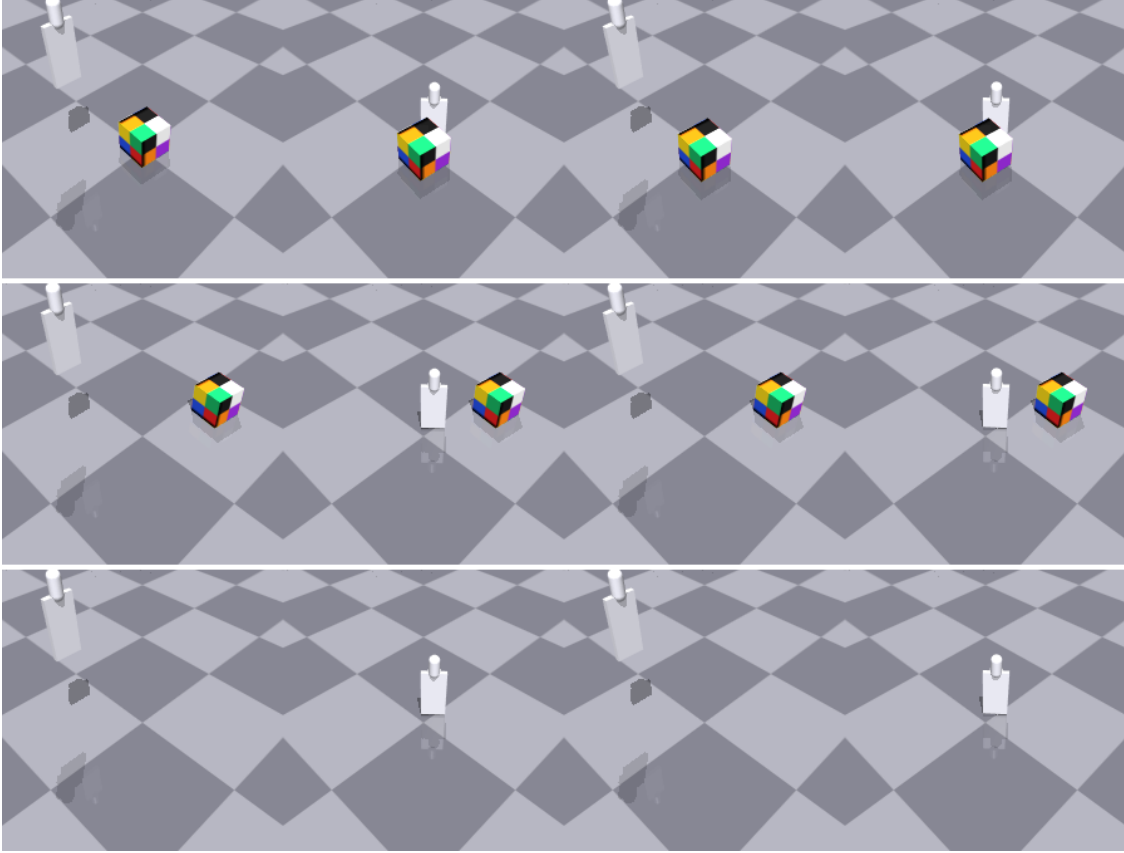


Figure 1: Real renders of the three conditions for one action. Rows, top to bottom: INTERACT, DECOY, ABSENT. Columns, left to right: s_0 , s_{std} , s_{del} , s_{time} . The arm motion is identical in every row. At s_{std} the arm is mid-push and encodes the action; at s_{del} it is back at rest and only the object’s settled pose carries the action; at s_{time} the arm has not retracted.

2 Setup

Task. A 4-DOF paddle pushes one of three objects (box, sphere, cylinder) in MuJoCo [16], rendered at 224×224 and 20 fps. The action $a = (v, \theta, d)$ (push speed 0.12–0.36 m/s, approach angle $\pm 90^\circ$, lateral contact offset ± 35 mm) is drawn uniformly. Each dimension is normalised to $[0, 1]$ and we report mean absolute error over the three dimensions, for which the uniform-prior baseline (“chance”) is 0.25. The arm is position-controlled and clamped to the commanded path, so its motion is identical across conditions; a yaw joint makes the map from action to arm pose invertible, the situation under study.

Conditions. For each (action, seed) tuple we simulate INTERACT (object in the push path), DECOY (the same object placed in an annulus verified numerically to clear every reachable push corridor by ≥ 14 mm, drawn from an independent random stream so its placement carries no information about the action) and ABSENT (no object). 2000 tuples per geometry give 18 000 rollouts. Splits are 80/10/10 by tuple, so paired rollouts never straddle them; the held-out set has 611 INTERACT/DECOY pairs after excluding rollouts that fail physics validators (at most 1.1% per cell).

Horizons (Fig. 1). s_{std} is the last frame of the push, the horizon the standard metric uses. $s_{\text{del}} = \max(\text{object settled, arm at rest})$. s_{time} has the same timing as s_{del} , but the arm only withdraws by 1.5 mm instead of retracting, which separates retraction from elapsed time.

Table 1: Main experiment (held-out, pooled over geometries; 611 pairs per seed, 5 seeds). Errors as % of chance (0.25). G relative to the reference error (= INTERACT column), and object share (fraction of the distance to chance removed by deleting the object’s physics). 95% CIs.

IDM	INTERACT	DECOY	ABSENT	G / reference	object share
A-std (standard)	1.9	2.2	1.5	0.15 [0.12, 0.19]	0.003
A-clip-del	2.2	2.5	2.0	0.14 [0.11, 0.16]	0.003
A-time	4.1	4.2	3.2	0.02 [−0.01, 0.06]	0.001
A-del (OG-AF)	44.7	99.0	98.5	1.21 [1.18, 1.24]	0.98

IDMs. We follow the MultiWorld protocol [17]: a ResNet-50 (ImageNet initialisation, our choice) with a non-causal temporal transformer, MSE loss. **A-std** reads a 16-frame clip spanning $[s_0, s_{\text{std}}]$, a faithful reproduction of the standard metric. **A-del** (OG-AF) reads only the pair (s_0, s_{del}) . **A-time** reads (s_0, s_{time}) , and **A-clip-del** reads a clip ending at s_{del} . Each has five seeds and is trained on all three conditions, so that no condition is out of distribution at test. Because the arm alone decodes the action (§3), training on INTERACT only would leave the same optimal solution available. As descriptive support we also train linear-probe-style MLPs on frozen DINOv3 ViT-L/16 CLS features [15] (ten seeds). DINOv3 resolves a one-frame object displacement at $+13.9\sigma$ where DINOv2 [14] reaches $+0.55\sigma$.

Estimator. On held-out tuples, $G = \text{mean}_p[\text{err}(\text{DECOY}_p) - \text{err}(\text{INTERACT}_p)]$, the mean of per-pair differences. A metric that uses the object should find DECOY much harder, giving a large G ; one that reads only the arm gives $G \approx 0$. For a physics-sensitive metric, *small G is the failure*. Absolute errors differ by $25\times$ across IDMs, so we report G relative to each IDM’s reference error, its own held-out INTERACT error (the INTERACT column of Table 1). We also report the *object share*, $G/(0.25 - \text{err}(\text{INTERACT}))$: the fraction of the IDM’s distance to chance that disappears when the object’s physics is removed. CIs are 95% hierarchical bootstrap (tuples within seeds, then seeds; 10 000 replicates).

Pre-registration. The confirmatory hypotheses, thresholds and estimators were committed to a public repository and time-stamped (2026-09-29) before the confirmatory experiment ran. A mechanical lock refuses to run that analysis until the registration is public and unmodified [**anonymised repository**]. The world-model test of §5 was registered separately (2026-10-05) before its corruption was calibrated or its model trained. Thresholds were fixed at pre-declared template defaults (for example $\tau_1 = 0.25$) without reference to any confirmatory statistic. A deviations log lists every post-registration change, and none affects a confirmatory result.

3 The standard metric ignores the object

State oracles. Probes on simulator state, before any pixels: arm state alone recovers the action to 1.0–1.2% of chance at s_{std} , identically with and without the object in the path. At s_{del} the arm carries nothing (97–98%) while the settled object carries 11–45% (original 3-seed record; an independent re-run gives 0.7–0.9%, 97–98% and 10–48%). The shortcut channel exists by construction. The question is whether trained pixel IDMs use anything else.

Confirmatory test 1: insensitivity of the standard metric. Deleting the object’s physics raises A-std’s error by $0.15\times$ its reference error, below the pre-registered threshold $\tau_1 = 0.25$ (CI upper bound 0.19; Holm-adjusted one-sided $p \leq 0.0002$, the bootstrap’s resolution). The gap is above zero, so the IDM uses the object slightly, but the object accounts for only 0.3% of A-std’s distance to chance. A-std is also *better* with no object at all (1.5% versus 1.9%), consistent with the object occluding or distracting from the arm.

Confirmatory test 2: OG-AF reads the object. At s_{del} OG-AF recovers the action from the settled object at 44.7% of chance (Holm $p \leq 0.0002$). As registered, this test is close to automatic, because it reduces

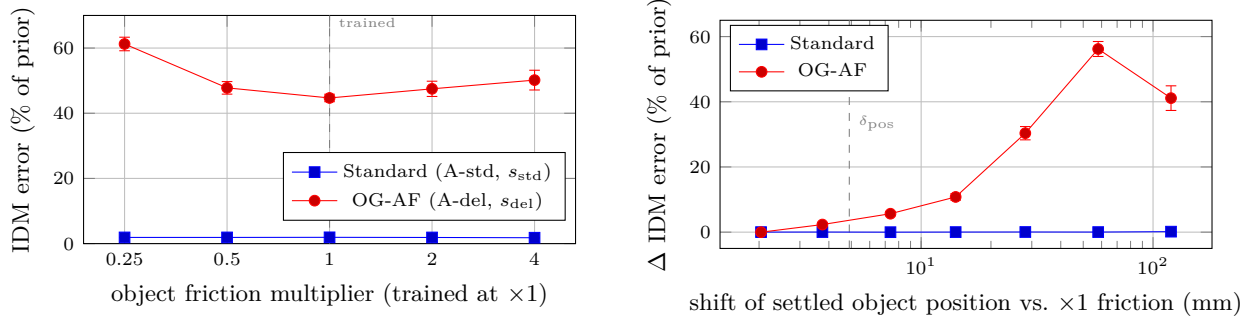


Figure 2: **Left:** IDMs trained at friction $\times 1$, scored on held-out rollouts with changed object friction (95% CI). The standard metric stays at 1.8–1.9% while the object comes to rest up to 6 cm away; OG-AF rises on both sides of the training value. **Right:** change in IDM error (percentage points of chance) against the physics-induced shift of the object’s resting position, binned. 1570 (tuple, friction) pairs over friction $\times \{0.1, 0.25, 0.5, 2, 3, 4\}$; $\delta_{pos} = 4.9$ mm is the 5th-percentile per-frame object displacement.

to “better than chance”. The informative number is the contrast with DECOY, where OG-AF is at chance (99.0%), as it must be: at s_{del} the arm is at rest in every condition. The object share is 0.98.

Descriptive checks. *Retraction, not elapsed time:* A-time behaves like A-std ($|\Delta G| = 0.0005$ in normalised error) and not A-del (0.135), so the effect comes from the arm retracting. *Clip access:* an IDM given a clip that ends at s_{del} (A-clip-del) re-opens the shortcut, with G matching A-std, so OG-AF must score a frame pair, not a clip. The frozen-feature probes show the same pattern ($G \approx 0$ at s_{std} , large at s_{del}).

4 How large a physics change each metric registers (exploratory)

Neither analysis in this section was pre-registered as a confirmatory test.

Friction sweep. We score the trained IDMs on rollouts whose object friction is changed from $\times 0.25$ to $\times 4$ (Fig. 2, left). The arm is identical across frictions, so the standard metric stays at 1.8–1.9% of chance, although the object comes to rest up to 6 cm away (median 20 mm at $\times 0.25$). OG-AF’s error rises away from the training friction in both directions: 61.2% at $\times 0.25$ and 50.2% at $\times 4$, versus 44.7% at $\times 1$. The differences at $\times 0.5$ and $\times 2$ are small. A prediction of a *monotone* response, in our protocol specification, was mis-specified for an IDM trained at $\times 1$, and is reported as failed.

Error change versus displacement. For each held-out tuple and friction level we pair the simulator’s shift of the object’s resting position with the change in each IDM’s error (Fig. 2, right). On the friction levels of the sweep alone, OG-AF’s error rises significantly from the 2.5–5 mm bin on (+2.1 points, 95% CI [+1.4, +2.9]), and the standard metric shows no significant change in any bin up to 40 mm. Adding the $\times 0.1$ and $\times 3$ corpora built for §5 extends the range: OG-AF’s paired effect grows from $d = 0.64$ at 2.5–5 mm to 3.27 at 40–80 mm. The standard metric’s only nonzero response, +0.15 points, appears beyond 80 mm and comes entirely from $\times 0.1$ rollouts. OG-AF’s response drops beyond 80 mm, plausibly as resting positions leave the range seen in training.

This flatness is a property of the design as much as of the metric: in a position-controlled setup, physics cannot move the arm, and the standard metric reads the arm. OG-AF’s reference error is higher (44.7% versus 1.9% of chance). It gives up precision on the action to gain sensitivity to the interaction, which is what an action-following metric is meant to measure.

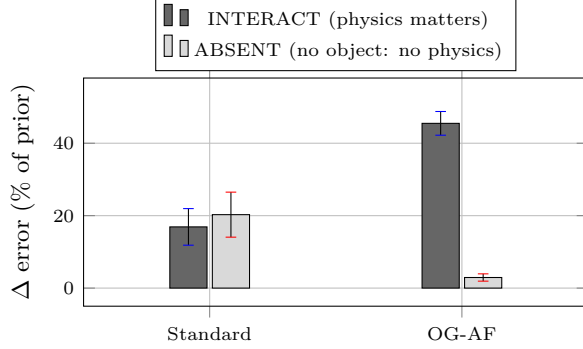


Figure 3: Change in mean IDM error, wrong-physics world model (friction $\times 0.1$) minus correct WM-base, on the same 193 held-out tuples, in percentage points of chance (95% paired-bootstrap CIs). The standard metric rises as much on ABSENT, where there is no object and physics cannot matter, as on INTERACT (difference -3.4 points, CI $[-8.3, +1.2]$). OG-AF rises by 45.5 points on INTERACT and 2.9 on ABSENT.

5 World models: catching wrong physics

Models. We train factorised spatio-temporal video transformers (90.5M parameters) over the $4 \times 28 \times 28$ latents of a compact convolutional VAE, with AdaLN action conditioning and strict temporal causality. Each is trained for 120k steps on INTERACT clips of all three geometries, 16 frames spanning $[s_0, s_{\text{del}}]$. Each model generates the 193 held-out box tuples deterministically and autoregressively from the real first frame. The registered ladder has WM-base at 100%, 30% and 10% of its training steps, a model trained on 10% of the data, and a model trained on wrong physics. Generated video is scored with IDMs trained with mild appearance augmentation (colour jitter, blur, compression artefacts; seed 0). Real-video numbers elsewhere use unaugmented IDMs averaged over five seeds.

Ranking agreement. On every model the standard and OG-AF metrics rank the same generated rollouts almost independently (Spearman $\rho \leq 0.28$). This is descriptive: the < 0.9 criterion in our analysis code is lax and does not appear in the registration.

A diagnosed failure. Our first ladder failed its validity check: models were not monotone in ground-truth error (Spearman 0.70, below the 0.9 our analysis requires), which voided the between-model claim as registered. The check measured whole-frame pixel error, which is dominated by how the arm and background are rendered (the object covers roughly 2% of the frame). The corruption (friction $\times 3$) was also never checked to be large. The validity check fell for the very shortcut it was meant to guard against.

The registered fix. Before running anything, we registered a second ladder. Its ground truth is object-region error: pixels where the real INTERACT and ABSENT frames differ. Its corruption is chosen mechanically from friction $\times \{0.1, 0.25, 4, 10\}$ as the mildest candidate whose *real* video already differs from correct physics by at least $1.25\times$ the weakest correct model’s object-region error, with at least 80% of rollouts passing the physics validators. Only $\times 0.1$ qualified. By intent, this makes the corrupted model’s position at the bottom of the ladder likely; whether models 1–4 are ordered correctly is not constrained.

Confirmatory test 3: the wrong-physics model. The rebuilt ladder is monotone in object-region error (Spearman 1.00; 0.057, 0.059, 0.070, 0.114, 0.162). Paired over the same tuples, the wrong-physics model differs from the correct one by $d = 1.94$ [1.73, 2.22] under OG-AF and $d = 0.46$ [0.37, 0.56] under the standard metric. The registered contrast is $\Delta d = 1.48$ [1.22, 1.79], which supports the hypothesis. The standard metric does respond, but its response is no larger on INTERACT than on the no-object control (Fig. 3). This is consistent with a rendering difference and not a physics one; the corrupted model also lost 12.6% of its training rollouts to the physics validators. For OG-AF, the difference between the INTERACT and ABSENT responses is +42.6 points [38.8, 46.1]. The corruption is gross: OG-AF scores the model near chance (92.4%).

Domain gap. On generated video the standard metric’s error is 56–57% of chance for the base models (1.9% on real video), and 61–64% on the ABSENT control. On generated video it largely measures rendering fidelity. OG-AF sits near its real-video reference (47% versus 44.7%) and near chance on ABSENT. Part of

this robustness is structural: half of OG-AF’s input is the real conditioning frame, and the arm at rest is easy to render. These numbers come from small self-trained models and may not transfer to larger ones.

6 Related work

Action-conditioned video world models for robotics [4, 7, 13, 18, 21] are increasingly evaluated by action following with IDMs [17], building on IDM-based action labelling [1, 19]. A recent diagnosis of whether robotic world models follow actions measures end-effector trajectories, i.e. arm kinematics, by design [5]. A position paper argues that decision-relevant evaluation, not visual fidelity, should guide world-model assessment [20]. Physical-plausibility benchmarks evaluate generated video directly [2, 3, 9, 12]; we study an evaluation pipeline already in use. The World Action Verifier [11] assumes a verification subset whose next state depends only on its own state and the action, and proves that a sparse inverse model transfers under that assumption. Read as an evaluation metric, the condition is object-blindness: in our data it holds at s_{std} and fails at s_{del} . The guarantee and the confound are the same property used for different purposes. More broadly, the embodiment shortcut is an instance of shortcut learning [8, 10] and of causal confusion [6].

7 Limitations

Scope. Simulation only: one push task, three primitive objects, a position-controlled arm whose trajectory contact cannot perturb, and small world models trained on their own corpus. On compliant or torque-controlled arms, contact perturbs the arm, so the arm channel will carry some physics signal, entangled with embodiment. The shortcut should shrink there but not vanish. Transfer to real robot data and to published world models is the main open question.

Generality of OG-AF. OG-AF needs a horizon at which the manipulator is at rest and the object has settled, which suits primitive-level actions. High-rate continuous control, grasping in which the arm stays in contact, and multi-step tasks would need extensions such as scoring at action-chunk boundaries, masking or inpainting the arm, or object tracking. Because OG-AF sees only the end state, it cannot detect wrong intermediate dynamics that end in the right state. Masking the arm at s_{std} is a natural alternative: in our frozen-feature probes, object pixels alone reach 48% of chance at s_{std} . Directly comparing tracked object trajectories with paired real video is another, where such video exists.

World-model evidence. One corruption type, which is gross; one IDM seed; 193 held-out box tuples (the registration anticipated 1 500, but the test split holds 193). The wrong-physics corpus lost 8–18% of rollouts per geometry to the physics validators, so that model also saw less data.

Other. DINO features are blind to sub-degree rotation. The analyses of §4 are exploratory. Two exploratory predictions failed as worded, the friction monotonicity and the orderings of a pixel-masking decomposition, and are reported as such.

8 Conclusion

Action following, as practised, rewards rendering the robot correctly. Scoring at a horizon where the manipulator no longer encodes the action turns the same pipeline into a measurement of the interaction. We recommend that action-following results report an object-grounded variant and a no-object control alongside the standard number.

Reproducibility. Code, both registration documents, the deviations log, every result record, and the generators for all tables and figures are in [anonymised repository]. One command resubmits the pipeline on a SLURM cluster. The full study used roughly 80 GPU-hours (A100/L40S), plus older GPUs for rendering.

Use of AI assistance. An AI coding agent (a large language model) implemented and audited substantial parts of the pipeline and analysis code, drafted parts of the manuscript, and, at the researchers’ delegation, fixed the open registration values: the template defaults, and the design of the world-model registration. It

did so before any confirmatory statistic existed, as recorded in both registration documents. The researchers reviewed the code, analyses and text and take responsibility for all decisions and claims.

References

- [1] Bowen Baker, Ilge Akkaya, Peter Zhokhov, Joost Huizinga, Jie Tang, Adrien Ecoffet, Brandon Houghton, Raul Sampedro, and Jeff Clune. Video pretraining (VPT): Learning to act by watching unlabeled online videos. In *Advances in Neural Information Processing Systems*, 2022.
- [2] Hritik Bansal, Zongyu Lin, Tianyi Xie, et al. VideoPhy: Evaluating physical commonsense for video generation. *arXiv preprint arXiv:2406.03520*, 2024.
- [3] Daniel M. Bear, Elias Wang, Damian Mrowca, et al. Physion: Evaluating physical prediction from vision in humans and machines. In *NeurIPS Datasets and Benchmarks Track*, 2021.
- [4] Jake Bruce, Michael Dennis, Ashley Edwards, et al. Genie: Generative interactive environments. In *International Conference on Machine Learning*, 2024.
- [5] Sixiang Chen, Jiaming Liu, Jixian Wu, Yichen Guo, Tinghao Wang, Siyuan Qian, Hao Chen, Jiajun Cao, Jian Tang, and Shanghang Zhang. Do robotic world models really follow actions? Diagnosing and aligning action-conditioned generation for policy learning. *arXiv preprint arXiv:2608.24885*, 2026.
- [6] Pim de Haan, Dinesh Jayaraman, and Sergey Levine. Causal confusion in imitation learning. In *Advances in Neural Information Processing Systems*, 2019.
- [7] Yilun Du, Mengjiao Yang, Bo Dai, Hanjun Dai, Ofir Nachum, Joshua B. Tenenbaum, Dale Schuurmans, and Pieter Abbeel. Learning universal policies via text-guided video generation. In *Advances in Neural Information Processing Systems*, 2023.
- [8] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2:665–673, 2020.
- [9] Bingyi Kang, Yang Yue, Rui Lu, Zhijie Lin, Yang Zhao, Kaixin Wang, Gao Huang, and Jiashi Feng. How far is video generation from world model: A physical law perspective. *arXiv preprint arXiv:2411.02385*, 2024.
- [10] Sebastian Lapuschkin, Stephan Wäldchen, Alexander Binder, Grégoire Montavon, Wojciech Samek, and Klaus-Robert Müller. Unmasking Clever Hans predictors and assessing what machines really learn. *Nature Communications*, 10, 2019.
- [11] Yuejiang Liu, Fan Feng, Lingjing Kong, Weifeng Lu, Jinzhou Tang, Kun Zhang, Kevin Murphy, Chelsea Finn, and Yilun Du. World action verifier: Self-improving world models via forward-inverse asymmetry. *arXiv preprint arXiv:2604.01985*, 2026.
- [12] Saman Motamed, Laura Culp, Kevin Swersky, Priyank Jaini, and Robert Geirhos. Do generative video models understand physical principles? *arXiv preprint arXiv:2501.09038*, 2025.
- [13] NVIDIA. Cosmos world foundation model platform for physical AI. *arXiv preprint arXiv:2501.03575*, 2025.
- [14] Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.
- [15] Oriane Siméoni et al. DINOv3. *arXiv preprint arXiv:2508.10104*, 2025.
- [16] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2012.
- [17] Haoyu Wu, Jiwen Yu, Yingtian Zou, and Xihui Liu. MultiWorld: Scalable multi-agent multi-view video world models. *arXiv preprint arXiv:2604.18564*, 2026.
- [18] Mengjiao Yang, Yilun Du, Kamyar Ghasemipour, Jonathan Tompson, Leslie Kaelbling, Dale Schuurmans, and Pieter Abbeel. Learning interactive real-world simulators. In *International Conference on Learning Representations*, 2024.

- [19] Seonghyeon Ye, Joel Jang, Byeongguk Jeon, et al. Latent action pretraining from videos. *arXiv preprint arXiv:2410.11758*, 2024.
- [20] Yang Yu, Shiyuan Zhang, Yifei Sheng, Haoxiang Ren, and Haoxin Lin. How should world models be evaluated for embodied decision-making? A decision-making-centric position. *arXiv preprint arXiv:2606.15032*, 2026.
- [21] Fangqi Zhu, Hongtao Wu, Song Guo, Yuxiao Liu, Chilam Cheang, and Tao Kong. IRASim: Learning interactive real-robot action simulators. *arXiv preprint arXiv:2406.14540*, 2024.